

Predictable Whole-Cell Kinetic Models with Default Enzyme Kinetics

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Abstract

Whole-cell kinetic models (WCKMs), based on Michaelis-Menten kinetics, require accurate kinetic parameters; such as, turnover numbers (kcat), and Michaelis-Menten constants (Km); which are often unavailable. To address this, many *ab initio* constructions of WCKMs or conversion from genome-scale metabolic models relies on default kcat of 13.7 per second and Km of 130 micromolar, which are based on average enzyme kinetics. However, the predictability of WCKMs on default kcat and Km values is unclear. This study evaluates the impact of such defaults on WCKMs. Our results show that local sensitivity analysis (LSA) using one-factor-at-a-time revealed that kcat is the significantly more sensitive (p-values < 0.0269) than Km; and the Pearson's correlation of time-course metabolite concentrations between using actual or default kcat values is 0.7166, suggesting substantial predictability despite using default kcat values. More importantly, fixing 5 % to 50% of the kcat values from default values to actual values (representing experimental verification of kcat values) show a monotonic improvement in predictability, where fixing 5% of reaction showed $r = 0.717$, and $r = 0.826$ at 50%. The findings imply that using default values provide a reasonable baseline but enzyme turnover numbers should be prioritised for experimental measurement during model construction as the benefit scales with the proportions of reactions validated.

Introduction

In metabolic engineering research, mathematical models help narrow down viable genetic modifications by offering predictive insights into pathway behaviour [1, 2]. Genome-scale models (GSMs) and whole cell kinetic models (WCKMs) dominate modelling efforts in this area [3, 4]. GSMs deliver reliable flux predictions but remain constrained to rate-level interpretations. WCKMs extend predictive power to both rates and yields [5], while also supporting gene knock-in simulations more efficiently [6]. This combination of predictive breadth and modelling flexibility makes WCKMs attractive for early-stage strategy evaluation. Consequently, an increasing number of researchers are calling for the development of additional kinetic models to support more accurate and versatile computational screening [7, 8].

Numerous GSMs have been developed [9], compared to WCKMs which are fewer in numbers [3]. This is due to the substantial data requirements and parameterization challenges associated with kinetic modelling at the whole-cell level. To address this gap, Amir-Hamzah et al. [10] had developed a means to convert GSMs into WCKMs using default enzyme kinetics [namely, turnover number (kcat) and Michaelis-Menten constant (Km)] on Michaelis-Menten styled approximation equations; which were found to be comparable to enzymological equations [11], such as Ping-Pong mechanism [12]. The default kcat of 13.7 per second and Km of 130 micromolar, used in Amir-Hamzah et al. [10] were median values from a survey of more than 1000 enzymes by Bar-Even et al. [13]. In turn, these default kcat and Km values have been used in many *ab initio* constructions of WCKMs [6, 14-20].

However, it is unclear to what extent such parameter approximations influence model accuracy, robustness, and reliability in guiding biological interpretation and engineering decisions. In this study, we investigate the effect of default enzyme kinetic parameters on the predictability of WCKMs. Our results suggest that WCKMs with default enzyme kinetic parameters may be predictable.

Methods

Sensitivity Analysis: Impact of Enzyme Kinetics and Initial Concentration. One-factor-at-a-time (OFAT) sensitivity analysis [21] was used to examine the impact of turnover number (kcat), Michaelis-Menten constant (km), and initial concentration on the metabolic fluxes of WCKMs. Two GSMs (e_coli_core containing 95 reactions with 72 metabolites [22] to represent a small sized GSM, and iT341 containing 554 reactions with 485 metabolites [23] to represent a medium sized GSM) were converted to their corresponding WCKMs using AdvanceSyn Toolkit with default kinetic parameters [10, 24]. OFAT sensitivity analysis [21] was performed on both converted models using AdvanceSyn Toolkit's local sensitivity analysis function and the endpoint concentration profiles were analyzed using Pearson's correlation against their respective GSM-converted WCKMs with default enzyme kinetics, where lower Pearson's correlation denotes higher sensitivity.

MultiBugs Experiment: Effects of Random Turnover Numbers. In this experiment, we examined the effects of random turnover numbers on the metabolic profiles (both time-course and endpoint metabolic concentration profiles). Only reactions containing both reactants and products from the WCKM of GSM-converted iT341 [23] were retained as DefaultBug, which consisted 477 reactions. We define ModelBug as DefaultBug with its original turnover numbers from iT341 [23]; thus, the basis of comparison. Then; 50 variants, denoted as TrialBug-1 to TrialBug-50; containing random variations for each turnover number for all reactions. The variations of turnover numbers were based on uniform distribution from 1 to 27 (± 13 from default turnover = 13.7, rounded up to 14). Pearson's correlation of time-course or endpoint metabolic concentration profiles of TrialBugs or DefaultBug against that of ModelBug was examined where high Pearson's correlation denotes better predictability.

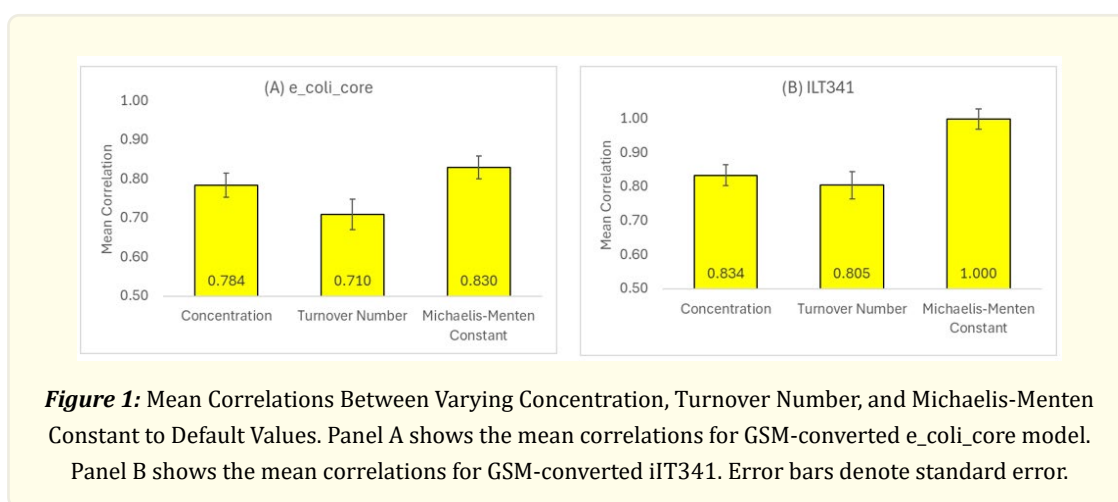
MetaBugs Experiment: Effects of Fixing Proportions of Turnover Numbers. In this experiment, we examine the effects of fixing of a proportion of turnover numbers from DefaultBug using the turnover numbers in ModelBug. This is to simulate the effects of experimentally validated turnover number on metabolic predictability. We then define a set of MetaBugs with varied proportions of fixing. MetaBug1 to MetaBug20 (n = 20) each had 5% of the turnover numbers fixed in sequential order; that is, MetaBug1 has first 5% of the reactions with turnover number from iT341 [23] while MetaBug2 has the second 5% of the reactions with turnover number from iT341 [23]. This resulted in reduced number of MetaBugs when the ratio of turnover number fixation increases. Therefore, MetaBug21 to MetaBug31 (n = 10), MetaBug31 to MetaBug35 (n = 5), MetaBug36 to MetaBug38 (n = 3), MetaBug39 and MetaBug40 (n = 2), and MetaBug41 and MetaBug42 (n = 2) had 10%, 20%, 30%, 40%, and 50% of the turnover numbers fixed. Similar to MultiBug experiment, Pearson's correlation of time-course or endpoint metabolic concentration profiles of MetaBugs against that of ModelBug was examined where high Pearson's correlation denotes better predictability.

Results and Discussion

Sensitivity Analysis Suggests that Turnover Numbers are More Sensitive than Michaelis-Menten Constants. Using OFAT sensitivity analysis [21] on both WCKMs converted from their respective GSMs, our results suggests that turnover number is significantly

more sensitive than Michaelis-Menten constant for both models (Figure 1; p-values < 0.0269). Using 2-samples t-test assuming unequal variance, the mean Pearson's correlation for varying turnover numbers is significantly lower than varying Michaelis-Menten constants (p-values < 0.0269) in GSM-converted *e_coli_core* [22].

Considering a larger model, GSM-converted iIT341 [23], the same significance holds (p-value < 2.64E-32). This suggests that turnover numbers are more sensitive than Michaelis-Menten constants, which is also supported by the view that turnover number is fundamental in understanding cellular metabolism [25], which resulted in more prediction models for turnover number than Michaelis-Menten constant [26]. Moreover, the mean Pearson's correlations for Michaelis-Menten constants are higher than that of initial concentrations in both cases. More importantly, sensitivity analysis of GSM-converted iIT341 [23] suggests that varying Michaelis-Menten constants marginal practical impact on endpoint concentration profiles as the mean Pearson's correlation is 1.000 when compared to default enzyme kinetics. Therefore, our work forward shall focus solely on turnover numbers.



Substantial Predictability from WCKMs with Default Turnover Number. Since sensitivity analysis of turnover numbers using GSM-converted iIT341 [23] (denoted as DefaultBug) suggests that the mean metabolic predictability of DefaultBug by means of Pearson's correlation is 0.805 (Figure 1B), this is only a suggestive value as a more accurate predictability requires comparing against the original turnover numbers found in iIT341 [23] (denoted as ModelBug). As such, the metabolic profile of ModelBug will be the basis of comparison.

Our results show that the correlations between the endpoint metabolic concentrations and time-course metabolic concentrations of DefaultBug and ModelBug are 0.7133 and 0.7166, respectively. The mean correlation coefficients between the endpoint metabolic concentrations and time-course metabolic concentrations of the 50 TrialBugs and ModelBug are 0.7219 ($t = 1.001$, p-value = 0.322) and 0.7214 ($t = 0.562$, p-value = 0.577), respectively (Figure 2A); and not significantly different to that between DefaultBug and ModelBug. Our results also show no significant differences ($n = 50$; paired t-test p-value = 0.121) by comparing the mean Pearson's correlation between the endpoint metabolic concentrations and time-course metabolic concentrations of the 50 TrialBugs against ModelBug (Figure 2B). This may suggest that the structure and size of the model, in terms of number of reactions and pathway complexity, may be an important factor in predictability; thus, providing an impetus for increasingly larger and more complex model constructions [27-29]. Nevertheless, our results suggests that default turnover number of 13.7 [13], which is used on the GSM to WCKM model converter [10] in AdvanceSyn Toolkit [24], provides substantial metabolic predictability.

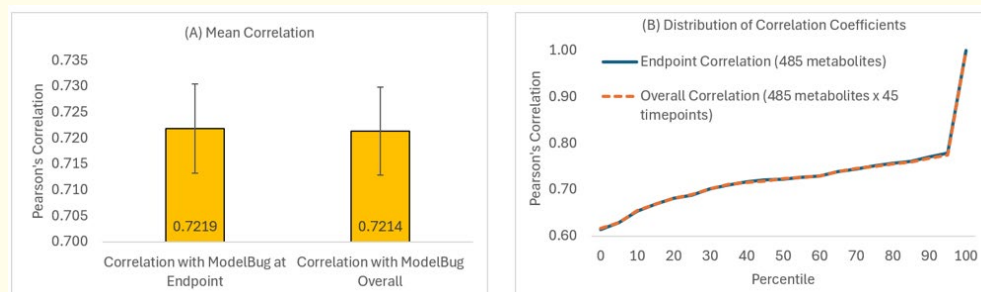


Figure 2: Predictability Between ModelBug and 50 TrialBugs.

Accurate Measurement of Some Turnover Numbers May Improve Predictability. If the default value of 13.7, which is the median turnover number [13], can provide substantial metabolic predictability based on either the endpoint metabolic or time-course metabolic concentrations (Figure 2A); then the question of whether experimental validation of turnover numbers can further improve metabolic predictability becomes focus. In fact, there is a call for sensitivity analysis as part of modelling and optimization [30]; when translated into the context of this study, can mean the identification and ranking of turnover numbers by their sensitivity to the resulting metabolic profiles [31]. However, the initial question remains - can experimental validation of the turnover numbers improve the predictability of metabolic profile?

Using the turnover numbers in ModelBug as basis, our results show an observable improvement in predictability in terms of both time-course or endpoint metabolic profiles by knowing (thereby, fixing) from 5 to 50 percent of the turnover numbers (Figure 3). This is supported by regression analysis showing that predictability of time-course ($F = 21.591$, $p\text{-value} = 3.62\text{E-}05$) or endpoint ($F = 22.774$, $p\text{-value} = 3.41\text{E-}05$) metabolic concentration profiles using percent fixing as regressor are significant. Given that 100% fixation of turnover numbers is equivalent to ModelBug itself; thereby, perfect predictability (Pearson's correlation = 1); our results suggest that experimental validation of turnover numbers has the potential to improve predictability over the default turnover number of 13.7 [10, 13].

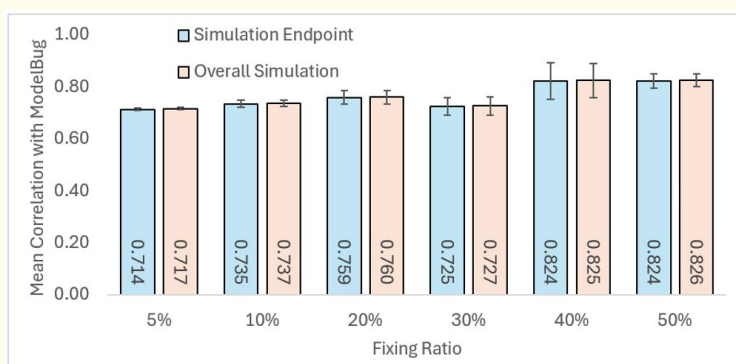


Figure 3: Predictability of Models by Simulating Accurate Measurements of Up to 50% of All Turnover Numbers.

Conclusion

This study investigates whether default turnover numbers and Michaelis-Menten constants are sufficient for accurate whole-cell kinetic simulations. Our results show that using default values can provide substantial predictability with the Pearson's correlation of 0.7166.

Supplementary Materials

Supplementary materials can be download from https://bit.ly/KM_predictability.

Conflict of Interest

The authors declare no conflict of interest.

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